

DMVPN Fase 2 con IKEv1 y OSPF

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Práctica: P4

Objetivo

Configurar una VPN hub and spoke punto a multipunto **DMVPN (Dynamic Multipoint VPN) Fase 2** con IKEv1 y enrutamiento dinámico OSPF entre un Hub y dos Spokes (Cisco CSR1000v), permitiendo comunicación cifrada tanto entre Hub↔Spoke como directamente entre Spoke↔Spoke sin pasar por el Hub.

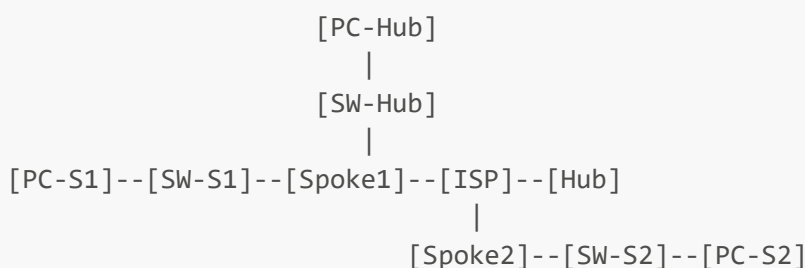
DMVPN combina tres tecnologías:

- **mGRE (Multipoint GRE):** Una sola interfaz Tunnel en el Hub acepta conexiones de múltiples Spokes simultáneamente.
- **NHRP (Next Hop Resolution Protocol):** Los Spokes se registran con el Hub como servidor NHS. En Fase 2, el Hub resuelve la IP NBMA (pública) del Spoke destino para que los Spokes puedan crear túneles directos entre sí.
- **IPSec:** Cifra todo el tráfico GRE que viaja por los túneles.

Diferencia entre Fases DMVPN

Característica	Fase 1	Fase 2	Fase 3
Tráfico Spoke↔Spoke	Pasa por Hub	Directo	Directo + summarization
NHRP redirect	No	Sí (Hub)	Sí (Hub)
NHRP shortcut	No	Sí (Spokes)	Sí (Spokes)
Routing en Hub	No split-horizon	No split-horizon	ip nhrp redirect
Escalabilidad	Baja	Media	Alta

Topología



Túneles DMVPN (mGRE sobre IPSec):
Hub (10.0.0.1) ↔ Spoke1 (10.0.0.2)

Hub (10.0.0.1) ↔ Spoke2 (10.0.0.3)
Spoke1 (10.0.0.2) ↔ Spoke2 (10.0.0.3) [túnel directo Fase 2]

Direccionamiento IP

Dispositivo	Interfaz	Dirección IP	Máscara	Descripción
Hub	GigabitEthernet1	25.7.73.193	255.255.255.252	WAN Hub → ISP
ISP	GigabitEthernet1	25.7.73.194	255.255.255.252	WAN ISP → Hub
ISP	GigabitEthernet2	25.7.73.197	255.255.255.252	WAN ISP → Spoke1
ISP	GigabitEthernet3	25.7.73.201	255.255.255.252	WAN ISP → Spoke2
Spoke1	GigabitEthernet2	25.7.73.198	255.255.255.252	WAN Spoke1 → ISP
Spoke2	GigabitEthernet2	25.7.73.202	255.255.255.252	WAN Spoke2 → ISP
Hub	GigabitEthernet3	25.7.73.205	255.255.255.248	Gateway LAN Hub
Spoke1	GigabitEthernet3	25.7.73.213	255.255.255.248	Gateway LAN Spoke1
Spoke2	GigabitEthernet3	25.7.73.221	255.255.255.248	Gateway LAN Spoke2
Hub	Tunnel0 (mGRE)	10.0.0.1	255.255.255.0	NHS (Hub DMVPN)
Spoke1	Tunnel0 (mGRE)	10.0.0.2	255.255.255.0	NHC Spoke1
Spoke2	Tunnel0 (mGRE)	10.0.0.3	255.255.255.0	NHC Spoke2
PC-Hub	eth0	25.7.73.206	255.255.255.248	Host LAN Hub
PC-S1	eth0	25.7.73.214	255.255.255.248	Host LAN Spoke1
PC-S2	eth0	25.7.73.222	255.255.255.248	Host LAN Spoke2

Subredes utilizadas (bloque 25.7.73.192/27)

Subred	Rango utilizable	Uso
25.7.73.192/30	25.7.73.193 – 25.7.73.194	WAN Hub – ISP
25.7.73.196/30	25.7.73.197 – 25.7.73.198	WAN ISP – Spoke1
25.7.73.200/30	25.7.73.201 – 25.7.73.202	WAN ISP – Spoke2
25.7.73.204/29	25.7.73.205 – 25.7.73.210	LAN Hub
25.7.73.208/29	25.7.73.209 – 25.7.73.214	LAN Spoke1
25.7.73.216/29	25.7.73.217 – 25.7.73.222	LAN Spoke2
10.0.0.0/24	10.0.0.1 – 10.0.0.254	Red DMVPN (Tunnel0)

La red del túnel DMVPN usa el espacio privado **10.0.0.0/24** para evitar solapamiento con el bloque **25.7.73.0/24** usado en las interfaces físicas.

Parámetros de configuración

IKEv1 ISAKMP Policy

Parámetro	Valor
Cifrado	AES 256
Hash	SHA-256
Autenticación	Pre-shared key
Grupo DH	Group 14 (2048-bit)
Lifetime	86400 segundos
Pre-shared key	Cisco123!
Peer address	0.0.0.0 (cualquier)

IPSec Transform Set

Parámetro	Valor
Cifrado ESP	AES 256
Integridad ESP	SHA-256 HMAC
Modo	Transport

Tunnel mGRE

Parámetro	Hub	Spoke1	Spoke2
IP Tunnel	10.0.0.1/24	10.0.0.2/24	10.0.0.3/24
Tunnel source	Gi1	Gi2	Gi2
Tunnel mode	gre multipoint	gre multipoint	gre multipoint
NHRP network-id	100	100	100
NHRP NHS	—	10.0.0.1	10.0.0.1
NHRP map al Hub	—	10.0.0.1→25.7.73.193	10.0.0.1→25.7.73.193
ip nhrp redirect	Sí	No	No
ip nhrp shortcut	No	Sí	Sí
ip ospf network	broadcast	broadcast	broadcast
ip ospf priority	2 (DR)	0 (never DR)	0 (never DR)

OSPF

Parámetro	Hub	Spoke1	Spoke2
Process ID	1	1	1
Area	0	0	0
Networks	10.0.0.0/24, 25.7.73.204/29	10.0.0.0/24, 25.7.73.208/29	10.0.0.0/24, 25.7.73.216/29

Scripts de configuración

ISP

```
enable
configure terminal
hostname ISP

interface GigabitEthernet1
 ip address 25.7.73.194 255.255.255.252
 no shutdown
 exit

interface GigabitEthernet2
 ip address 25.7.73.197 255.255.255.252
 no shutdown
 exit

interface GigabitEthernet3
 ip address 25.7.73.201 255.255.255.252
 no shutdown
 exit

end
write memory
```

Hub

```
enable
configure terminal
hostname Hub

interface GigabitEthernet1
 ip address 25.7.73.193 255.255.255.252
 no shutdown
 exit

interface GigabitEthernet3
 ip address 25.7.73.205 255.255.255.248
 no shutdown
```

```
exit

! --- Ruta default y rutas hacia WANs de Spokes ---
ip route 0.0.0.0 0.0.0.0 25.7.73.194
ip route 25.7.73.196 255.255.255.252 25.7.73.194
ip route 25.7.73.200 255.255.255.252 25.7.73.194

! --- IKEv1 ---
crypto isakmp policy 10
  encryption aes 256
  hash sha256
  authentication pre-share
  group 14
  lifetime 86400
exit

crypto isakmp key Cisco123! address 0.0.0.0 0.0.0.0

crypto ipsec transform-set TS-DMVPN esp-aes 256 esp-sha256-hmac
  mode transport
exit

crypto ipsec profile PROF-DMVPN
  set transform-set TS-DMVPN
exit

! --- Tunnel mGRE (Hub) ---
interface Tunnel0
  ip address 10.0.0.1 255.255.255.0
  ip nhrp authentication Cisco123
  ip nhrp map multicast dynamic
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp redirect
  ip ospf network broadcast
  ip ospf priority 2
  tunnel source GigabitEthernet1
  tunnel mode gre multipoint
  tunnel protection ipsec profile PROF-DMVPN
  no shutdown
exit

! --- OSPF ---
router ospf 1
  network 10.0.0.0 0.0.0.255 area 0
  network 25.7.73.204 0.0.0.7 area 0
exit

end
write memory
```

Spoke1

```
enable
configure terminal
hostname Spoke1

interface GigabitEthernet2
 ip address 25.7.73.198 255.255.255.252
 no shutdown
 exit

interface GigabitEthernet3
 ip address 25.7.73.213 255.255.255.248
 no shutdown
 exit

! --- Ruta default y rutas explícitas hacia WAN de Spoke2 ---
ip route 0.0.0.0 0.0.0.0 25.7.73.197
ip route 25.7.73.200 255.255.255.252 25.7.73.197
ip route 25.7.73.202 255.255.255.255 25.7.73.197

! --- IKEv1 ---
crypto isakmp policy 10
 encryption aes 256
 hash sha256
 authentication pre-share
 group 14
 lifetime 86400
 exit

crypto isakmp key Cisco123! address 0.0.0.0 0.0.0.0

crypto ipsec transform-set TS-DMVPN esp-aes 256 esp-sha256-hmac
 mode transport
 exit

crypto ipsec profile PROF-DMVPN
 set transform-set TS-DMVPN
 exit

! --- Tunnel mGRE (Spoke) ---
interface Tunnel0
 ip address 10.0.0.2 255.255.255.0
 ip nhrp authentication Cisco123
 ip nhrp map 10.0.0.1 25.7.73.193
 ip nhrp map multicast 25.7.73.193
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 10.0.0.1
 ip nhrp shortcut
 ip ospf network broadcast
 ip ospf priority 0
 tunnel source GigabitEthernet2
 tunnel mode gre multipoint
```

```
tunnel protection ipsec profile PROF-DMVPN
no shutdown
exit

! --- OSPF ---
router ospf 1
 network 10.0.0.0 0.0.0.255 area 0
 network 25.7.73.208 0.0.0.7 area 0
exit

end
write memory
```

Spoke2

```
enable
configure terminal
hostname Spoke2

interface GigabitEthernet2
 ip address 25.7.73.202 255.255.255.252
 no shutdown
 exit

interface GigabitEthernet3
 ip address 25.7.73.221 255.255.255.248
 no shutdown
 exit

! --- Ruta default y rutas explícitas hacia WAN de Spoke1 ---
ip route 0.0.0.0 0.0.0.0 25.7.73.201
ip route 25.7.73.196 255.255.255.252 25.7.73.201
ip route 25.7.73.198 255.255.255.255 25.7.73.201

! --- IKEv1 ---
crypto isakmp policy 10
 encryption aes 256
 hash sha256
 authentication pre-share
 group 14
 lifetime 86400
 exit

crypto isakmp key Cisco123! address 0.0.0.0 0.0.0.0

crypto ipsec transform-set TS-DMVPN esp-aes 256 esp-sha256-hmac
 mode transport
 exit

crypto ipsec profile PROF-DMVPN
 set transform-set TS-DMVPN
```

```
exit

! --- Tunnel mGRE (Spoke) ---
interface Tunnel0
 ip address 10.0.0.3 255.255.255.0
 ip nhrp authentication Cisco123
 ip nhrp map 10.0.0.1 25.7.73.193
 ip nhrp map multicast 25.7.73.193
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 10.0.0.1
 ip nhrp shortcut
 ip ospf network broadcast
 ip ospf priority 0
 tunnel source GigabitEthernet2
 tunnel mode gre multipoint
 tunnel protection ipsec profile PROF-DMVPN
 no shutdown
 exit

! --- OSPF ---
router ospf 1
 network 10.0.0.0 0.0.0.255 area 0
 network 25.7.73.216 0.0.0.7 area 0
 exit

end
write memory
```

SW-Hub, SW-S1, SW-S2 (IOSvL2)

```
enable
configure terminal

interface GigabitEthernet0/0
 switchport mode access
 switchport access vlan 1
 no shutdown
 exit

interface GigabitEthernet0/1
 switchport mode access
 switchport access vlan 1
 no shutdown
 exit

end
write memory
```

PC-Hub, PC-S1, PC-S2 (VPCS)


```
! PC-Hub
ip 25.7.73.206 255.255.255.248 25.7.73.205
save

! PC-S1
ip 25.7.73.214 255.255.255.248 25.7.73.213
save

! PC-S2
ip 25.7.73.222 255.255.255.248 25.7.73.221
save
```

Verificación y pruebas

Comandos de verificación ejecutados en Hub, Spoke1 y Spoke2.

Comandos de verificación

```
! En Hub
show dmvpn
show ip ospf neighbor
show crypto isakmp sa
show ip route ospf

! En Spoke1 y Spoke2
show dmvpn
show crypto isakmp sa
```

Resultados obtenidos — Hub

show dmvpn:

```
Type:Hub, NHRP Peers:2
25.7.73.198 10.0.0.2 UP Dynamic ← Spoke1 registrado
25.7.73.202 10.0.0.3 UP Dynamic ← Spoke2 registrado
```

show ip ospf neighbor:

```
25.7.73.213 0 FULL/DROTHER Tunnel0 ← Spoke1
25.7.73.221 0 FULL/DROTHER Tunnel0 ← Spoke2
```

show ip route ospf:

```
0 25.7.73.208/29 [110/1001] via 10.0.0.2, Tunnel0 ← LAN Spoke1
0 25.7.73.216/29 [110/1001] via 10.0.0.3, Tunnel0 ← LAN Spoke2
```

Resultados obtenidos — Spoke1

show dmvpn:

```
Type:Spoke, NHRP Peers:2
25.7.73.193 10.0.0.1 UP Static ← Hub (entrada estática)
25.7.73.202 10.0.0.3 UP Dynamic ← Spoke2 (túnel directo Fase 2)
```

show crypto isakmp sa:

```
25.7.73.198 25.7.73.202 QM_IDLE ACTIVE ← SA directa Spoke1↔Spoke2
25.7.73.193 25.7.73.198 QM_IDLE ACTIVE ← SA Spoke1↔Hub
```

Pruebas de conectividad

Hub → Spokes:

```
Hub# ping 25.7.73.213
Success rate is 100 percent (5/5)

Hub# ping 25.7.73.221
Success rate is 100 percent (5/5)
```

Spoke1 → Spoke2 (túnel directo):

```
Spoke1# ping 25.7.73.221 source GigabitEthernet3
! Primer intento: 0% (negociación NHRP + IPSec)
! Segundo intento: 100% (túnel directo establecido)
Success rate is 100 percent (5/5)
```

Spoke2 → Spoke1 (túnel directo):

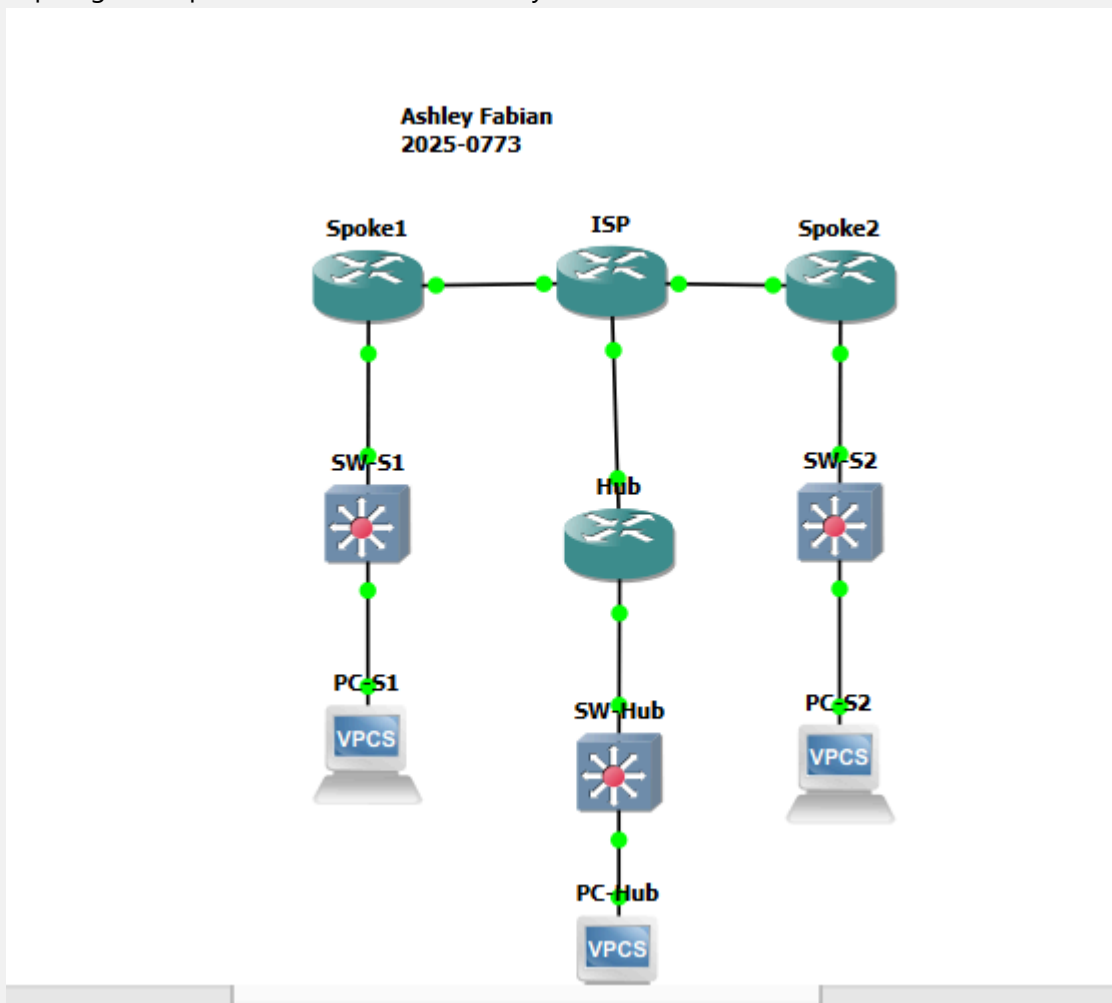
```
Spoke2# ping 25.7.73.213 source GigabitEthernet3
Success rate is 100 percent (5/5)
```

El primer ping entre Spokes falla mientras NHRP resuelve la IP NBMA del Spoke destino y se negocia la SA IPSec directa. Este es el comportamiento esperado y definitorio de DMVPN Fase 2.

Capturas de pantalla

Insertar aquí las capturas de:

1. Topología completa en GNS3 con nombre y matrícula visible



2. `show dmvpn` en Hub (2 peers UP Dynamic)

```
Hub#show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel

=====
Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
  1 25.7.73.198          10.0.0.2    UP 01:28:41    D
  1 25.7.73.202          10.0.0.3    UP 01:01:02    D

Hub#
```

solarwinds | Solar-PuTTY free tool

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7. show crypto isakmp sa en Spoke1 (SA directa con Spoke2)

```
Spoke1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id status
25.7.73.193  25.7.73.198  QM_IDLE       1001  ACTIVE

IPv6 Crypto ISAKMP SA
Spoke1#
```

8. Ping Spoke1 → Spoke2 (primer fallo + segundo exitoso)

```
Spoke1#ping 25.7.73.221 source Gi3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 25.7.73.221, timeout is 2 seconds:
Packet sent with a source address of 25.7.73.213
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 30/43/80 ms
Spoke1#
```

9. Ping Spoke2 → Spoke1 exitoso

```
Spoke2#ping 25.7.73.213 source Gi3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 25.7.73.213, timeout is 2 seconds:
Packet sent with a source address of 25.7.73.221
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 17/34/58 ms
Spoke2#
```

Conclusión

Se configuró exitosamente DMVPN Fase 2 con IKEv1 y OSPF entre un Hub y dos Spokes. Los Spokes se registraron dinámicamente con el Hub mediante NHRP, y OSPF distribuyó las rutas de las LANs a través del túnel mGRE. La característica definitoria de la Fase 2 — el túnel directo Spoke↔Spoke sin pasar por el Hub — fue verificada mediante el comportamiento del primer ping fallido (durante la resolución NHRP y negociación IPsec) seguido de pings exitosos al establecerse la SA directa entre Spoke1 (25.7.73.198) y Spoke2 (25.7.73.202). Las dos SAs IKEv1 en cada Spoke (una hacia el Hub y otra hacia el Spoke remoto) confirmaron el funcionamiento correcto de la Fase 2.